

بسم الله الرحمن الرحيم

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1. Run and analyze the "Classifying Handwritten Digits with MLP Network" example. Extract the dimensions of the arrays and analyze them using equations.
2. Design an MLP network for the classification of the Heart Disease Detection dataset (CSV file attached to HW2).
 - a. You may reference the following links for guidance, but you must design and train your own unique network.
 - b. Develop an appropriate and optimal training and testing routine.
 - c. If necessary, apply preprocessing steps to the data to achieve better results.
 - d. Analyze the various sections of the provided code.
 - e. Provide at least four insightful analyses of the obtained results. Use appropriate figures and graphs to illustrate your analyses and employ different visualization methods to display and interpret the results effectively.
 - * f. Achieving high accuracy and comparing the results with existing methods can earn you an extra score. Aim to demonstrate that your method outperforms previous approaches, and compare the performance of your network with existing results on this dataset.

<https://github.com/anik199/Heart-failure-prediction/blob/main/heart-failure-beginner-friendly-91-accuracy.ipynb>

<https://github.com/sahilverma0696/heart-disease-prediction/blob/master/main.ipynb>

<https://github.com/Miker2808/Heart-Disease-Detection-FFNN/tree/main>

1. Design an MLP network for the regression application of the Boston dataset, applying all sections outlined in Question 2.

<https://github.com/christianversloot/machine-learning-articles/blob/main/how-to-create-a-neural-network-for-regression-with-pytorch.md>

```
from sklearn.datasets import load_boston
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موفق باشید.